



ALCOHOL — FULL MIND MAP

Chemistry · Chapter 7.B · Topic-wise · Fully Explained · Hinglish · SSC/PCS/UPSC Ready

Topics in this chapter:

1. Methanol vs Ethanol & Denatured Alcohol
2. Fermentation of Sugar & Enzymes
3. Molasses, Barley & Alcoholic Beverages
4. Breathalyzer Chemistry

1. Methanol vs Ethanol & Denatured Alcohol

- Ethyl alcohol (ethanol) ko drinking ke liye unfit banane ke context mein source answer Methanol + Pyridine diya gaya hai. Aise treated ethanol ko denatured alcohol/methylated spirit category mein rakha jata hai. [Q1]
- Methanol (methyl alcohol) bahut toxic hota hai; poisoning se metabolic acidosis, optic-nerve damage, blindness aur severe cases mein death tak ho sakti hai. Liquor tragedies mein methanol contamination classic cause hai. [Q2]
- Methyl alcohol ka formula CH_3OH hai aur ise Wood Spirit bhi kaha jata hai. Historical preparation wood ki destructive distillation se hoti thi. [Q3]



EXAM TRAP — Denatured alcohol ko “methanol + pyridine hi hamesha” mat samjho

- ▶ Source/old exam answer Q1 mein Methanol + Pyridine hai; practical denaturing formulations country/regulation ke hisab se vary kar sakti hain.
- ▶ Ethanol khud beverage alcohol hai; Methanol toxic alcohol hai—dono ko interchange mat karo.

2. Fermentation of Sugar & Enzymes

- Sugar/glucose ki fermentation se Ethyl alcohol (Ethanol) banta hai. Ye process yeast ke enzymes ke through anaerobic conditions mein hota hai. [Q4]
- Glucose ko ethanol + CO_2 mein convert karne wala enzyme-complex Zymase hai. Yeast mein naturally present hota hai aur alcoholic fermentation mein central role play karta hai. [Q5]
- Enzyme recall: Invertase → sucrose ko glucose + fructose; Maltase → maltose ko glucose; Diastase/Amylase-type action → starch ko simpler sugars/maltose mein todta hai. [Q5]

3. Molasses, Barley & Alcoholic Beverages

- Molasses (sheera) sugar industry ka by-product hai aur fermentation ke through alcohol/ethanol production ka important raw material hai. Rum traditionally molasses route se ban sakti hai. [Q6]
- Alcoholic beverages ke raw materials alag ho sakte hain: Beer mainly malted barley; Brandy grapes; Whisky barley/corn/grain; Cider apple; Champagne grapes. Source mein alcohol-% ranges bhi indicative form mein diye gaye hain. [Q6]
- Germinated barley se malt milta hai aur uske fermentation se Beer banayi jati hai. Isliye given options mein correct answer Beer hai. [Q7]



EXAM TRAP — Beverage source ko rigid one-to-one rule mat banao

- ▶ Whisky, beer etc. ki production traditions/raw materials region aur style ke hisab se vary kar sakte hain; exam question ka specific option-context dekho.
- ▶ Q7 ka key clue “germinated barley grains → malt fermentation” hai, jiska standard answer Beer hai.

4. Breathalyzer Chemistry

- Old chemical breathalyzer mein acidified Potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4$) alcohol vapour ke saath react karta tha. Ethanol oxidise hota hai aur dichromate ka orange colour green Cr^{3+} side ki taraf change hota hai. [Q8]
- Colour change ki degree breath alcohol level se relate ki jati thi; isi principle par source ka filter-paper question based hai. [Q8]



EXAM TRAP — Old breathalyzer vs modern devices

- ▶ Q8 specifically old dichromate-based breathalyzer chemistry poochta hai.
- ▶ Modern breath alcohol analysers commonly fuel-cell sensor ya infrared spectroscopy bhi use kar sakte hain; isliye “har breathalyzer mein dichromate filter paper hota hai” mat likho.

✓ Source-question mapping preserved: Q1–Q8